

Programmable stigmatic singlet lenses

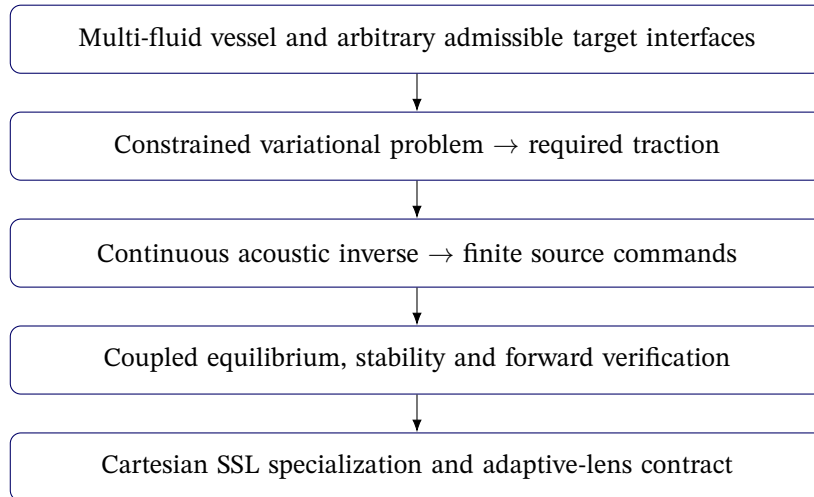
A fixed-prescription acoustic compiler
Vision, theoretical framework, and verification contract

Acoustic Freeform Lab
Private research design report

24 September 2026 — Version 1.2: multi-fluid variational foundation

Abstract

We start from a vessel containing multiple immiscible fluids and a prescribed collection of admissible three-dimensional interfaces. A constrained shape Lagrangian identifies the required mechanical traction as the multiplier of the desired geometry, with pressure gauge freedom determined by compartment volumes and connectivity. A viscous Rayleighian extends this inverse to compatible interface velocities; an inertial weak formulation states its limits. Only after establishing this load problem do we introduce continuous acoustic sources, finite-array amplitude and phase optimization, and coupled forward verification. Cartesian singlets are a later target specialization with user-supplied conjugates, thickness and frame radius. Stationary accuracy, stability, formation and adaptive transitions are distinct claims. This is a mathematical design specification, not a universal realizability theorem or an experimental validation.



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Part I

Mathematical formulation and derivations

1 The multi-fluid problem for a prescribed shape

The starting point is a desired collection of fluid interfaces, not an optical prescription or an emitter array. The two questions are: *which mechanical tractions support the prescribed shape, and which admissible sources can realize those tractions?* Cartesian lenses are a later specialization of the target geometry.

1.1 Domains, interfaces and admissible data

A fixed bounded vessel $\mathcal{D} \subset \mathbb{R}^3$ contains L connected fluid regions $\Omega_\ell(\Gamma)$ separated by J smooth interfaces $\Gamma = (\Gamma_1, \dots, \Gamma_J)$. On each interface choose a normal $\mathbf{n}_j$ from region ℓ_j^- to region ℓ_j^+ . Define

$$[\rho]_j^- = \rho_{\ell_j^-} - \rho_{\ell_j^+}, \quad [T]_j = T_{\ell_j^+} - T_{\ell_j^-}, \quad \kappa_j = \nabla_{\Gamma_j} \cdot \mathbf{n}_j. \quad (1.1)$$

Thus an outward normal on a spherical inner phase has $\kappa = 2/R_s$. The two jump conventions are written explicitly to avoid mixing density differences with stress jumps. Distinct disconnected regions of the same material count as distinct compartments for pressure and volume constraints.

For the baseline formulation assume immiscible incompressible Newtonian fluids with constant $\rho_\ell > 0$, $\mu_\ell > 0$, constant $\sigma_j > 0$, and gravitational potential $\Phi(\mathbf{x}) = gz$. Interfaces may be fully three-dimensional; axisymmetry is not imposed. The vessel walls are impermeable and stationary, and contact lines are pinned. Topology is fixed, interfaces remain separated, and no triple-line or moving-contact-line law is introduced. Such laws, compressibility, thermal evolution and mass transfer require additional state equations and are not hidden in this baseline.

The prescribed target Γ^* must satisfy support, topology, clearance and phase-volume conditions. Let the independent conserved quantities be

$$C_a(\Gamma) = \sum_{\ell=1}^L c_{a\ell} |\Omega_\ell(\Gamma)| = C_a^0, \quad a = 1, \dots, m. \quad (1.2)$$

Usually these select $L-1$ independent compartment volumes, since total vessel volume is fixed. Reservoirs or bypass connections change the constraint set and must be part of the apparatus specification. A target violating Eq. (1.2) cannot be corrected by adjusting pressure multipliers.

1.2 Forward multi-fluid equations before a static reduction

Let $\mathbf{U}_\ell, p_\ell$ be mean velocity and physical pressure, and $T_\ell = -p_\ell I + 2\mu_\ell D(\mathbf{U}_\ell)$, where $D(\mathbf{U}) = (\nabla \mathbf{U} + \nabla \mathbf{U}^T)/2$. The bulk equations are

$$\nabla \cdot \mathbf{U}_\ell = 0, \quad (1.3)$$

$$\rho_\ell(\partial_t \mathbf{U}_\ell + \mathbf{U}_\ell \cdot \nabla \mathbf{U}_\ell) = \nabla \cdot T_\ell - \rho_\ell \nabla \Phi + \mathbf{b}_\ell. \quad (1.4)$$

Here $\mathbf{b}_\ell$ denotes separately specified mean bulk forcing. For clean interfaces with continuous velocity, a supplied normal load f_j with units Pa is defined by

$$V_{n,j} = \mathbf{U}^- \cdot \mathbf{n}_j = \mathbf{U}^+ \cdot \mathbf{n}_j, \quad [T]_j \mathbf{n}_j = (\sigma_j \kappa_j - f_j) \mathbf{n}_j. \quad (1.5)$$

Tangential stresses are continuous in this scalar-load baseline. Tangential actuation or variable surface tension changes this condition. The sign of f_j is chosen so its power input is $\int_{\Gamma_j} f_j V_{n,j} dS$. For zero load and a static sphere this convention recovers $p^- - p^+ = 2\sigma/R_s$.

These are slow/mean equations. If fast acoustic motion is resolved directly in a compressible parent model, radiation forces must not be added a second time. The acoustic sections below instead specify a scale-separated harmonic model that supplies f_j and states its omitted mechanisms.

2 The constrained variational inverse for desired interfaces

2.1 Energy and admissible shape variations

For the static load inversion in this section set $\mathbf{U} = 0$ and $\mathbf{b}_\ell = 0$: the only bulk body force is gravity. Physical pressure then has the hydrostatic form $p_\ell = p_\ell^0 - \rho_\ell \Phi$. Nonconservative bulk forcing or a streaming equilibrium instead requires the full weak balance developed below. For constant surface tension, define the energy

$$E(\Gamma) = \sum_{j=1}^J \sigma_j |\Gamma_j| + \sum_{\ell=1}^L \rho_\ell \int_{\Omega_\ell(\Gamma)} \Phi dV. \quad (2.1)$$

Fixed wall energies do not vary for the stated pinned contact lines. If η_j is normal interface displacement, then

$$DE(\Gamma)[\eta] = \sum_j \int_{\Gamma_j} (\sigma_j \kappa_j + [\rho]_j^- \Phi) \eta_j dS, \quad (2.2)$$

$$DC_a(\Gamma)[\eta] = \sum_j B_{aj} \int_{\Gamma_j} \eta_j dS, \quad B_{aj} = c_{a\ell_j^-} - c_{a\ell_j^+}. \quad (2.3)$$

The first identity follows from the area variation $\delta|\Gamma_j| = \int \kappa_j \eta_j dS$ and the transport theorem: moving the interface along $\mathbf{n}_j$ adds volume to its minus side and removes it from its plus side. Boundary terms vanish for pinned admissible variations. Equation (2.3) keeps the actual compartment connectivity in the inverse problem.

2.2 The desired shape is a constraint; its multiplier is traction

Use a local normal chart about a smooth target,

$$\Gamma_j(\chi_j) = \{\mathbf{X} + \chi_j(\mathbf{X}) \mathbf{n}_j^*(\mathbf{X}) : \mathbf{X} \in \Gamma_j^*\}, \quad \chi_j = 0 \text{ at the pinned rim.} \quad (2.4)$$

The target is $\chi = 0$; the chart is restricted to regular, nonintersecting interfaces. The geometric variational problem is to find a stationary point in χ and multipliers (λ, Λ) of

$$\mathcal{L}(\chi, \lambda, \Lambda) = E(\Gamma(\chi)) - \sum_a \lambda_a (C_a(\Gamma(\chi)) - C_a^0) - \sum_j \int_{\Gamma_j^*} \Lambda_j \chi_j dS^*. \quad (2.5)$$

This is a constrained stationarity problem, not an assumption that the energy is convex or that a global minimum exists. The geometric multiplier Λ_j is a field, in Pa; a scalar could not impose a pointwise shape. Its virtual work is evaluated on the target chart, which suffices for the first variation at $\chi = 0$.

Variations in the three groups of variables yield

$$\delta_\Lambda \mathcal{L} = 0 \implies \chi_j = 0, \quad (2.6)$$

$$\partial_{\lambda_a} \mathcal{L} = 0 \implies C_a(\Gamma^*) = C_a^0, \quad (2.7)$$

$$\delta_\chi \mathcal{L} = 0 \implies \Lambda_j = \sigma_j \kappa_j^* + [\rho]_j^- \Phi - \sum_a B_{aj} \lambda_a. \quad (2.8)$$

Consequently $\Lambda_j = f_{\text{req},j}$ is the normal traction that would support the target in the quiescent model. This is the required mechanical load, not a prescribed oscillatory acoustic pressure. No source has yet been shown to produce it.

Proposition 2.1 (Static load inversion). *For a regular compatible target, any source-generated normal load equal to Eq. (2.8), with the stated volume, boundary and quiescent bulk conditions, satisfies the constrained mechanical first variation at that target. Conversely, any such equilibrium satisfies this identity modulo its independent volume multipliers.*

Proof. Substitute Eq. (2.8) into $DE[\eta] - \sum_j \int f_j \eta_j \, dS$. The remainder is $\sum_a \lambda_a DC_a[\eta] = 0$ on admissible variations. Conversely, under independence of the volume differentials, a force functional that vanishes on their common kernel lies in their finite-dimensional span. Applying Eqs. (2.2) and (2.3) gives the stated representation. $\square$

Prescribing the complete target makes its volume constraints redundant in the augmented problem. The resulting nonuniqueness is physical pressure gauge freedom: if $\lambda \mapsto \lambda + \delta\lambda$, then $\Lambda_j \mapsto \Lambda_j - \sum_a B_{aj} \delta\lambda_a$. Only the pressure-difference subspace allowed by B may be removed. Two isolated exterior compartments permit different constants from a single connected bath. Subtracting an independent mean on every interface without examining this topology can incorrectly remove a required load.

2.3 Partial prescriptions and genuine free boundaries

If only subsets $\Gamma_{j,o}^*$ are prescribed, restrict the last integral in Eq. (2.5) to those subsets. On an unprescribed part with no applied load, stationarity gives

$$\sigma_j \kappa_j + [\rho]_j^- \Phi - \sum_a B_{aj} \lambda_a = 0. \quad (2.9)$$

That is a particular energy-based completion policy, not a universal choice of non-optical annulus. If other tractions act there, they enter its right side; if another completion functional is chosen, its Euler equation must be stated. Smooth joining conditions exclude hidden concentrated line forces. The fixed-prescription SSL version requires complete target interfaces or an explicitly specified completion before acoustic inversion.

3 A viscous variational problem for prescribed motion

The preceding inverse concerns equilibrium. A compatible desired normal velocity v_d requires an additional load. At the current geometry define

$$a_\Gamma(\mathbf{U}, \mathbf{W}) = \sum_\ell \int_{\Omega_\ell} 2\mu_\ell D(\mathbf{U}) : D(\mathbf{W}) \, dV, \quad (3.1)$$

$$(G_\Gamma \mathbf{U})_j = \mathbf{U} \cdot \mathbf{n}_j, \quad \langle f, v \rangle_\Gamma = \sum_j \int_{\Gamma_j} f_j v_j \, dS. \quad (3.2)$$

The velocity space has continuous interfacial traces and no-slip walls. In the Stokes limit, with no additional mean bulk force, consider the Rayleighian saddle functional

$$\mathcal{R}(\mathbf{U}, \pi, f) = \frac{1}{2}a_\Gamma(\mathbf{U}, \mathbf{U}) + DE(\Gamma)[G_\Gamma \mathbf{U}] - \sum_\ell \int_{\Omega_\ell} \pi_\ell \nabla \cdot \mathbf{U}_\ell dV - \langle f, G_\Gamma \mathbf{U} - v_d \rangle_\Gamma. \quad (3.3)$$

Here $DE[G\mathbf{U}]$ is energy rate, so every term has units W. The multiplier $\pi_\ell = p_\ell + \rho_\ell \Phi$ is reduced pressure; using the gravitational shape energy and also adding gravity work again would double count it. The variational dissipation construction is related to [1]; our pinned interfaces do not adopt its moving-contact-line law.

The saddle equations are

$$\nabla \cdot \mathbf{U}_\ell = 0, \quad G_\Gamma \mathbf{U} = v_d, \quad (3.4)$$

$$a_\Gamma(\mathbf{U}, \mathbf{W}) + DE[G_\Gamma \mathbf{W}] - \sum_\ell \int \pi_\ell \nabla \cdot \mathbf{W}_\ell dV = \langle f, G_\Gamma \mathbf{W} \rangle_\Gamma \quad \forall \mathbf{W}. \quad (3.5)$$

The requested velocity must conserve every sealed volume: $\sum_j B_{aj} \int_{\Gamma_j} v_{d,j} dS = 0$. Under appropriate regularity and compatibility, define the resistance by the constrained dissipation minimum

$$\frac{1}{2} \langle v, Z_\Gamma v \rangle_\Gamma = \min_{\nabla \cdot \mathbf{U}=0, G_\Gamma \mathbf{U}=v} \frac{1}{2} a_\Gamma(\mathbf{U}, \mathbf{U}). \quad (3.6)$$

This definition supplies symmetry and nonnegative dissipation; strict positivity/invertibility additionally require exclusion of null motions. With pressure constants understood on the volume-constrained space,

$$\boxed{f_{\text{req}} = E'_\Gamma + Z_\Gamma v_d - B^T \lambda, \quad v = M_\Gamma(f - E'_\Gamma), \quad M_\Gamma = Z_\Gamma^{-1}.} \quad (3.7)$$

At $v_d = 0$ this reduces to Eq. (2.8). Off-diagonal resistance blocks couple different interfaces through shared fluid regions; independent scalar relaxation laws for each face do not reproduce this variational problem.

With inertia or additional mean forcing, use the weak momentum balance instead of asserting the Stokes Rayleighian remains complete:

$$\begin{aligned} \langle f - E'_\Gamma, G_\Gamma \mathbf{W} \rangle_\Gamma &= \sum_\ell \int \rho_\ell D_t \mathbf{U}_\ell \cdot \mathbf{W}_\ell dV + a_\Gamma(\mathbf{U}, \mathbf{W}) \\ &\quad - \sum_\ell \int \pi_\ell \nabla \cdot \mathbf{W}_\ell dV - \mathcal{W}_{\text{additional}}(\mathbf{W}). \end{aligned} \quad (3.8)$$

For normal pressure alone to realize a desired motion, this right-hand functional must vanish on divergence-free test fields in $\ker G_\Gamma$ and must define a bounded functional on the normal trace space. Otherwise it cannot factor through normal interface work. This necessary compatibility condition makes clear that arbitrary fluid motion is not controllable with a scalar normal load. In a streaming base state, stationary interfaces do not imply $\mathbf{U} = 0$, so the quiescent formula is insufficient.

4 From the variational load to realizable sources

The geometric multipliers determine what is needed, not what is reachable. Let $\mathcal{U}_{\text{ad}}$ be the allowed continuous source space and let $\mathcal{Q}_{\Gamma^*}(u)$ be the cycle-mean normal traction generated by a declared wave model on the target. The exact stationary inverse is

$$\mathcal{Q}_{\Gamma^*}(u) = E'_{\Gamma^*} - B^T \lambda, \quad u \in \mathcal{U}_{\text{ad}}, \quad (4.1)$$

together with tangential/bulk compatibility and any power or field limits. A useful regularized formulation is

$$\min_{u \in \mathcal{U}_{\text{ad}}, \lambda} \frac{1}{2} \|Q_{\Gamma^*}(u) - E'_{\Gamma^*} + B^T \lambda\|_{\mathcal{Y}}^2 + \beta \mathcal{R}(u). \quad (4.2)$$

The norm $\mathcal{Y}$ and regularizer use explicit reference scales. Minimizing over λ projects out only the legitimate volume-pressure subspace. A positive residual may reflect lack of authority, discretization or local optimization failure; it is not by itself an impossibility certificate.

The logical sequence is therefore

$$\Gamma^* \longrightarrow (E', B, \Lambda) \longrightarrow \mathcal{Q}_{\Gamma^*}^{-1}(\Lambda) \longrightarrow \text{coupled forward verification}. \quad (4.3)$$

The inverse notation denotes a possibly empty, nonunique preimage, not an assumed invertible operator. Finite emitters restrict $\mathcal{U}_{\text{ad}}$; they do not define the underlying variational problem. The following sections specialize its mechanics to layered graphs, derive the acoustic source map and its optimality equations, and only then construct Cartesian SSL targets.

5 Layered-graph specialization of the variational load

Now specialize the preceding three-dimensional formulation to three sealed layers in a cylinder. Write their two complete interfaces as absolute graphs $z = h_0(x, y)$ and $z = h_1(x, y)$ over the horizontal disk D_R of radius R . Here $j = 0, 1$ labels interfaces and $\ell = 0, 1, 2$ labels fluids; $dA = dx dy$. The two graph integrals are fixed by the two independent compartment volumes. All horizontal variations are allowed; radial formulas are introduced only where explicitly stated. No optical shape has yet been prescribed.

5.1 Energy and hydrostatic pressure multipliers

With upward graph normals, let $\Delta\rho_j = \rho_j - \rho_{j+1}$ and take gravity to be $-g\mathbf{e}_z$. Direct integration of $\rho g z$ in the three phases yields, up to fixed end-cap terms,

$$E[h] = \sum_{j=0}^1 \int_{D_R} \left[\sigma_j \sqrt{1 + |\nabla h_j|^2} + \frac{1}{2} \Delta\rho_j g h_j^2 \right] dA. \quad (5.1)$$

Here h_j are absolute heights. Shifting the reference planes changes constant load terms that must be absorbed consistently into the volume multipliers. It does not remove gravity.

Varying the area term and integrating by parts with $\eta_j = 0$ at the rim,

$$DE[h](\eta) = \sum_j \int_{D_R} \left(\sigma_j \frac{\nabla h_j \cdot \nabla \eta_j}{\sqrt{1 + |\nabla h_j|^2}} + \Delta\rho_j g h_j \eta_j \right) dA \quad (5.2)$$

$$= \sum_j \int_{D_R} (\sigma_j \kappa_j + \Delta\rho_j g h_j) \eta_j dA, \quad (5.3)$$

$$\kappa_j = -\nabla \cdot \frac{\nabla h_j}{\sqrt{1 + |\nabla h_j|^2}}. \quad (5.4)$$

For a scalar upward normal traction f_j , its virtual work under the graph displacement $\eta_j \mathbf{e}_z$ is $\int_{\Gamma_j} f_j n_z \eta_j dS = \int_{D_R} f_j \eta_j dA$, since $n_z dS = dA$. Thus no extra area factor belongs in the graph load.

Stationarity under all constrained variations is equivalent to

$$\boxed{f_j^*(r) = \sigma_j \kappa_j[h^*](r) + \Delta\rho_j g h_j^*(r) - \lambda_j, \quad j = 0, 1.} \quad (5.5)$$

The constants λ_j have units Pa and are fixed-volume pressure multipliers; there are two independent constants, not freely adjustable spatial functions. The load is defined modulo these constants in the volume-constrained inverse. For axisymmetric graphs,

$$\kappa_j = -\frac{h_j''}{(1 + h_j'^2)^{3/2}} - \frac{h_j'}{r\sqrt{1 + h_j'^2}}, \quad \kappa_j(0) = -2h_j''(0). \quad (5.6)$$

An upper spherical cap of radius a_s has $\kappa = 2/a_s$, recovering the Laplace-pressure sign. A horizontal unforced graph has zero curvature and is balanced by a constant hydrostatic multiplier.

5.2 Second variation and the importance of the gravity sign

For admissible variations η, ξ ,

$$D^2 E[h](\eta, \xi) = \sum_j \int_{D_R} \left[\sigma_j \nabla \eta_j^T \left(\frac{I}{s_j} - \frac{\nabla h_j \nabla h_j^T}{s_j^3} \right) \nabla \xi_j + \Delta\rho_j g \eta_j \xi_j \right] dA, \quad s_j = \sqrt{1 + |\nabla h_j|^2}. \quad (5.7)$$

The axisymmetric capillary integrand reduces to $\sigma_j \eta_j' \xi_j' / (1 + h_j'^2)^{3/2}$. At a flat interface, let λ_1 be the infimum of $\int |\nabla \eta|^2 / \int |\eta|^2$ over its nonzero pinned zero-volume variations. A sufficient mechanical coercivity condition is

$$\sigma_j \lambda_1 + \Delta \rho_j g > 0 \quad (j = 0, 1). \quad (5.8)$$

Negative $\Delta \rho_j$ is destabilizing, but capillarity and confinement may offset it. This condition concerns the unforced energy, not stability under geometry-dependent acoustic loading.

6 Forward acoustics: equations, weak form and power

6.1 Declared linear harmonic reduction

For peak phasors proportional to $e^{-i\omega t}$, set $\alpha_\ell = 1/\rho_\ell$ and $\beta_\ell = \omega^2/(\rho_\ell c_\ell^2)$. In each homogeneous inviscid phase,

$$-\nabla \cdot (\alpha_\ell \nabla P_\ell) - \beta_\ell P_\ell = 0, \quad \mathbf{V}_\ell = \frac{\alpha_\ell \nabla P_\ell}{i\omega}. \quad (6.1)$$

In the high-frequency interface reduction used here,

$$P_j = P_{j+1}, \quad \alpha_j \partial_n P_j = \alpha_{j+1} \partial_n P_{j+1}, \quad (6.2)$$

with the same upward normal on both sides. At the domain-outward wall,

$$\mathbf{V} \cdot \mathbf{n}_a = YP + u, \quad \alpha \partial_n P = i\omega(YP + u), \quad \text{Re } Y \geq 0. \quad (6.3)$$

The command u is in m/s, Y in m/(Pa s), and P in Pa. If $Y \neq 0$, u is not total wall velocity. No electrical calibration is needed to define this idealized mathematical source.

Multiply the Helmholtz equation by a conjugate test pressure $\bar{\psi}$, integrate by parts and cancel the two-sided interface fluxes. Then

$$a_h(P, \psi) = \sum_\ell \int_{\Omega_\ell(h)} (\alpha_\ell \nabla P_\ell \cdot \nabla \bar{\psi} - \beta_\ell P_\ell \bar{\psi}) dV - i\omega \int_{\partial\Omega} Y P \bar{\psi} dS, \quad (6.4)$$

$$b(u, \psi) = i\omega \int_{\Gamma_a} u \bar{\psi} dS, \quad a_h(P, \psi) = b(u, \psi). \quad (6.5)$$

The inverse $\mathcal{A}_h^{-1}$ is used only where this wave problem is well posed. An undamped cavity resonance cannot simply be assigned a finite inverse response. Boundary conditions, resonance and regularization matter in active Helmholtz control [2].

6.2 Independent energy identity

Taking the imaginary part with $\psi = P$ in the weak equation gives

$$P_{\text{src}} = -\frac{1}{2} \text{Re} \int_{\Gamma_a} P \bar{u} dS = \frac{1}{2} \int_{\partial\Omega} \text{Re } Y |P|^2 dS. \quad (6.6)$$

The minus sign is from the outward velocity convention. This power is in W; it is not electrical power or the squared Euclidean command norm. Bulk dissipation and mean-flow energy transfer require additional terms if those mechanisms are included. The present identity is for the declared frozen, lossless-bulk harmonic problem.

7 From the acoustic field to the two mean tractions

In a homogeneous inviscid phase the normal component of the averaged acoustic momentum-flux tensor is

$$\mathcal{M}_{nn} = \frac{|P|^2}{4\rho c^2} - \frac{\rho}{4}|\mathbf{V}|^2 + \frac{\rho}{2}|V_n|^2 = \frac{|P|^2}{4\rho c^2} + \frac{\rho}{4}(|V_n|^2 - |\mathbf{V}_t|^2). \quad (7.1)$$

The graph-compatible upward traction is the lower-minus-upper jump,

$$f_{ac,j} = \mathcal{M}_{nn,j} - \mathcal{M}_{nn,j+1}. \quad (7.2)$$

Write v_n for the common normal velocity, and $G_t = \nabla_\Gamma P$ for the tangential pressure gradient. Using the transmission conditions gives

$$f_{ac,j} = \frac{1}{4} \left(\frac{1}{\rho_j c_j^2} - \frac{1}{\rho_{j+1} c_{j+1}^2} \right) |P|^2 + \frac{\rho_j - \rho_{j+1}}{4} |v_n|^2 + \frac{\rho_j - \rho_{j+1}}{4\omega^2 \rho_j \rho_{j+1}} |G_t|^2. \quad (7.3)$$

All three terms are in Pa. In particular, there is not one universal positive intensity-to-load factor at a two-fluid interface. In the pressure-release, negligible-upper-fluid limit, $P = G_t = 0$ and the remaining traction is $\rho|v_n|^2/4$. Wave-driven interface deformation and feedback of geometry on the field are studied in [3].

Under these ideal transmission conditions the tangential momentum-flux jump cancels because $\rho\mathbf{V}_t = G_t/(i\omega)$ and v_n are common. This does not eliminate boundary-layer streaming or thermal stresses in a more complete physical model. Their closures change the mean state and must be added consistently, not inferred from a passive scalar admittance; see [4, 5].

8 Continuous inverse, finite sources and optimality

8.1 The source space precedes the emitter array

Let $\mathcal{U}$ be a complex Hilbert space of admissible boundary commands on Γ_a , with explicit physical scales in its norm. Let S_h map commands to the pressure/velocity traces. Assume sufficient regularity for the quadratic traction pairing to be bounded. For a dimensionless admissible test function ϕ_i on the two graphs, define

$$F_i(u; h) = \sum_j \int_{D_R} \phi_{ij} f_{ac,j}(u; h) \, dA = \langle u, \mathcal{K}_i(h)u \rangle, \quad (8.1)$$

$$b_i(h^*) = DE[h^*](\phi_i), \quad \mathcal{K}_i = S_h^* C_i S_h. \quad (8.2)$$

The operators are Hermitian but not necessarily positive. Exact quiescent normal balance requires $F_i = b_i$ for a dense set of pinned, zero-volume tests, plus the remaining model conditions. Matching only pupil tests does not enforce full-interface equilibrium.

Choose dimensionless source patterns e_k and restrict

$$u(x) = \sum_{k=1}^N e_k(x) w_k, \quad w_k = A_k e^{i\varphi_k}, \quad [A_k] = \text{m/s}. \quad (8.3)$$

If H is the wave-trace response matrix, the finite force matrices are $K_i = H^\dagger C_i H$ and

$$F_i(w) = w^\dagger K_i w = \sum_{k,l} \bar{w}_k (K_i)_{kl} w_l, \quad [K_i] = \text{kg/m}. \quad (8.4)$$

The cross terms are the interference responsible for phase control. A sum of independent intensities is a different source model.

8.2 Differential and constrained source equations

At fixed target geometry,

$$DF_i(w)[\delta w] = 2 \operatorname{Re}[(K_i w)^\dagger \delta w], \quad J_i = 2((\operatorname{Re} K_i w)^T, (\operatorname{Im} K_i w)^T), \quad (8.5)$$

where $x = (\operatorname{Re} w, \operatorname{Im} w) \in \mathbb{R}^{2N}$. In amplitude/phase coordinates,

$$\frac{\partial F_i}{\partial A_k} = 2 \operatorname{Re}[(\overline{K_i w})_k e^{i\varphi_k}], \quad \frac{\partial F_i}{\partial \varphi_k} = -2 \operatorname{Im}[(\overline{K_i w})_k w_k]. \quad (8.6)$$

Real/imaginary coordinates avoid the singular phase parameterization at $A_k = 0$. A common phase change leaves every force unchanged, giving $J(-\operatorname{Im} w, \operatorname{Re} w)^T = 0$.

Let $s_i > 0$ be force scales, $D_s = \operatorname{diag}(s_i)$, $r_i = (F_i - b_i)/s_i$, and L a scaled real source regularizer. A version-1 objective and disk constraints are

$$\min_x \Phi(x) = \frac{1}{2} \|r(x)\|_2^2 + \frac{\beta}{2} \|Lx\|_2^2, \quad c_k(x) = x_k^2 + x_{N+k}^2 - w_{k,\max}^2 \leq 0. \quad (8.7)$$

For a differentiable local minimizer satisfying a constraint qualification, the necessary Karush–Kuhn–Tucker equations are

$$J^T D_s^{-1} r + \beta L^T Lx + \sum_k \mu_k \nabla c_k = 0, \quad (8.8)$$

$$c_k \leq 0, \quad \mu_k \geq 0, \quad \mu_k c_k = 0. \quad (8.9)$$

They are not a global optimality certificate for this nonconvex problem. At zero command the quadratic-load derivative vanishes, so initialization and nonzero finite-amplitude search are material algorithmic choices.

For $W = ww^\dagger$, exact coherent synthesis is equivalently constrained by

$$\operatorname{tr}(K_i W) = b_i, \quad W \succeq 0, \quad \operatorname{rank} W \leq 1, \quad W_{kk} \leq w_{k,\max}^2. \quad (8.10)$$

Dropping rank yields a semidefinite relaxation, not a realizable deterministic drive unless a rank-one command is recovered and rechecked. For any real a_i and $\|w\|_2 \leq C_w$, necessary reachability conditions include

$$\left| \sum_i a_i b_i \right| \leq C_w^2 \left\| \sum_i a_i K_i \right\|_2, \quad \sum_i a_i K_i \succeq 0 \implies \sum_i a_i b_i \geq 0. \quad (8.11)$$

A violation proves infeasibility for that model and bound; optimizer failure alone does not. Adding source patterns enlarges the feasible set only when the old patterns and their bounds remain available.

9 Coupled equilibrium, sensitivities and accuracy

9.1 The physical residual versus a numerical root iteration

Expand graphs about a compatible reference using dimensionless pinned, volume-preserving basis functions, $h = h_{\text{ref}} + \sum q_i \phi_i$. Coordinates q_i are in m, and the residual components are in N:

$$R(q, x) = \nabla_q E(q) - F_{\text{ac}}(q, x) = 0. \quad (9.1)$$

Each residual evaluation requires acoustics on the *current* geometry. A Newton step solves

$$K_{\text{eff}} \Delta q = -R, \quad K_{\text{eff}} = K_{\text{mech}} - A_{\text{ac}}, \quad K_{\text{mech}} = \nabla_q^2 E, \quad A_{\text{ac}} = \partial_q F_{\text{ac}}. \quad (9.2)$$

Line search or trust regions enforce geometry and residual safeguards. The iteration number is not physical time. A target-based initial guess is legitimate, but a converged root does not select a physical basin of attraction.

The acoustic shape derivative includes the changed wave, not only the evaluation location. Differentiating $A(q)P(q, x) = B(q)w(x)$ gives

$$A \delta P = B \delta w + (\delta B)w - (\delta A)P. \quad (9.3)$$

Substitute this into the differentiated quadratic traction and include normal, area and basis changes where applicable. Frozen wave sensitivities are not this full derivative. Check the assembled derivative independently with finite differences at several step sizes.

9.2 Static command sensitivity and coupled adjoint

On a regular equilibrium branch define $B_{ac} = \partial_x F_{ac}$. Then

$$K_{eff} \delta q = B_{ac} \delta x, \quad \frac{\partial q}{\partial x} = K_{eff}^{-1} B_{ac}. \quad (9.4)$$

For a smooth real objective $\mathcal{J}(q, x)$, the adjoint and reduced gradient are

$$K_{eff}^T \lambda = \partial_q \mathcal{J}, \quad \frac{d\mathcal{J}}{dx} = \partial_x \mathcal{J} + B_{ac}^T \lambda. \quad (9.5)$$

This follows directly by eliminating δq in the total differential; see the general adjoint framework [6]. Clipping, branch changes and maxima attained by changing rays are nonsmooth events and need separate constraint treatment, not blind use of this formula.

9.3 From force mismatch to shape and spot error

At the target q^* , let $r_\star = R(q^*, x)$ and let a nearby actual root be $q^* + \delta q$. Locally,

$$\delta q = -K_{eff}^{-1} r_\star + O(\|\delta q\|^2). \quad (9.6)$$

This expansion requires regularity and proximity; an ill-conditioned inverse can amplify small pressure mismatch. With scaled physical norms and a height-evaluation operator H_j ,

$$e_{h,j} \lesssim \|H_j K_{eff}^{-1}\| \|r_\star\|, \quad e_{h,j} = \sup_{0 \leq r \leq a_j} |h_j(r) - h_j^*(r)|. \quad (9.7)$$

The symbol $\lesssim$ denotes a leading-order diagnostic here, not a rigorous global error bound. A purely mechanical inverse K_{mech}^{-1} cannot replace K_{eff}^{-1} without justifying neglected acoustic feedback.

Fix the launch cone $\mathcal{L}$ and detector $z = z_2$. If a ray exits at $\mathbf{x}$ in direction $\mathbf{d}$ with $d_z > 0$, its detector coordinate is

$$\mathbf{s} = \mathbf{x}_\perp + \frac{z_2 - x_z}{d_z} \mathbf{d}_\perp, \quad e_{spot} = \sup_{\ell \in \mathcal{L}} \|\mathbf{s}(\ell)\|_2. \quad (9.8)$$

Each interface intersection solves

$$x_z + \tau d_z - h_j(|\mathbf{x}_\perp + \tau \mathbf{d}_\perp|) = 0, \quad \tau > 0, \quad (9.9)$$

followed by the Snell map in Eq. (12.5). Reject rays that fail interception, transmission, pupil or physical-apparatus clearance; do not remove them from the denominator of an acceptance statement. For a smooth fully transmitted ray family, $\delta \mathbf{s} = S_q \delta q$ to first order. Thus local optical sensitivity involves $S_q K_{eff}^{-1}$, not pressure error alone and not height error alone without slope information.

The joint stationary accuracy test is

$$e_{h,0}, e_{h,1} \leq \varepsilon_h, \quad e_{\text{spot}} \leq \varepsilon_s, \quad \text{all prescribed rays transmitted}, \quad (9.10)$$

with fixed coordinates, no piston removal and no best-focus shift. For this project $\varepsilon_h = 10^{-8}$ m. A strict $\varepsilon_s = 5 \times 10^{-7}$ m must not be confused with approximate acceptance of a 5.13×10^{-7} m result. Sampled maxima require their sampling and refinement evidence; they are not automatic continuum certificates.

10 Stability, formation and adaptable SSL families

10.1 Variational Stokes evolution and its limits

For incompressible three-phase mean velocities $\mathbf{U}$, define

$$a_h(\mathbf{U}, \mathbf{W}) = \sum_{\ell} \int_{\Omega_{\ell}(h)} 2\mu_{\ell} D(\mathbf{U}) : D(\mathbf{W}) \, dV, \quad (G_h \mathbf{U})_j = U_z - \nabla h_j \cdot \mathbf{U}_{\perp}. \quad (10.1)$$

Velocities are continuous across interfaces and zero on no-slip walls. Minimizing $a_h(\mathbf{U}, \mathbf{U})/2$ subject to incompressibility and $G_h \mathbf{U} = \dot{h}$ defines a symmetric resistance Z_h where the constrained problem is well posed. Its inverse is the Stokes mobility M_h . The reduced graph balance and evolution are

$$Z_h \dot{h} = f_{\text{ac}}(h, u) - E'_h, \quad \dot{q} = M(q)[F_{\text{ac}}(q, x(t)) - \nabla_q E(q)]. \quad (10.2)$$

For a fixed imposed load f^* independent of geometry,

$$\frac{d}{dt}(E[h] - \langle f^*, h \rangle) = -\langle \dot{h}, Z_h \dot{h} \rangle \leq 0. \quad (10.3)$$

This identity does not apply unchanged to a geometry-dependent acoustic load; fixed emitter commands are not fixed traction. The distinction is essential when interpreting a prescribed-pressure formation test.

At a quiescent equilibrium with fixed commands, $R = 0$ removes the derivative of the mobility from the linearization:

$$\delta \dot{q} = -M_{\star} K_{\text{eff}} \delta q. \quad (10.4)$$

In a finite-dimensional Stokes model, eigenvalues strictly in the left half-plane imply local linear asymptotic stability. A positive mechanical Hessian alone is insufficient, and nonnormality can permit transient growth.

An inertial model must retain fluid momentum and its coupling consistently:

$$\rho_{\ell}(\partial_t \mathbf{U}_{\ell} + \mathbf{U}_{\ell} \cdot \nabla \mathbf{U}_{\ell}) = -\nabla \bar{p}_{\ell} + \mu_{\ell} \Delta \mathbf{U}_{\ell} - \rho_{\ell} g \mathbf{e}_z + \mathbf{f}_{\ell, \text{additional}}, \quad \nabla \cdot \mathbf{U}_{\ell} = 0. \quad (10.5)$$

The acoustic interface traction enters the mean stress condition; extra bulk forces must be derived without double counting. A justified finite reduction may produce $M_{\text{in}} \delta \ddot{q} + D \delta \dot{q} + K_{\text{eff}} \delta q = 0$, but that equation is not a substitute for deriving its operators.

10.2 Adaptive commands and fixed-machine constraints

Let a parameter vector p describe requested optical states, with user-supplied z_1 . A differentiable admissible path must satisfy

$$c(p; \mathcal{M}) = 0, \quad D_p c \dot{p} = 0. \quad (10.6)$$

For exact equilibrium tracking, $R(q^*(p), x(p)) = 0$ gives the necessary local source-authority equation

$$B_{ac}\dot{x} = K_{eff} \frac{\partial q^*}{\partial p} \dot{p}. \quad (10.7)$$

It must be solvable within the tangent cone of active command limits. This relation tests local authority; it does not prescribe continuation as the inverse method or prove existence of a global path. At finite speed, the Stokes load would additionally require

$$F_{ac}(q_d, x(t)) = \nabla E(q_d) + Z(q_d)\dot{q}_d, \quad (10.8)$$

and an inertial/streaming model adds its corresponding stresses. This explains why equilibrium commands alone need not preserve stigmatism during motion.

Axisymmetric design does not exclude perturbations proportional to $e^{im\theta}$ with $m \neq 0$. Stability of a streaming base flow requires linearizing that flow, acoustic coupling and boundary conditions; a quiescent spectrum cannot certify it. Thermal feedback, changing indices and curing belong to separate declared extensions.

11 Cartesian singlets as a prescribed target family

The objective is to determine source commands for a *prescribed* lens, not to optimize its conjugates. The fixed prescription is

$$\mathcal{P} = (n_0, n_1, n_2, z_0, z_1, z_2, v_0, t, R, a, \lambda_{opt}), \quad v_1 = v_0 + t, \quad 0 < a \leq R, \quad t > 0. \quad (11.1)$$

All axial positions are laboratory coordinates. The user supplies z_1 ; t is central thickness, R the bounding radius, and a the qualified optical radius. The two target graphs h_j^* satisfy

$$\Gamma_j = \{(r \cos \theta, r \sin \theta, h_j(r)) : 0 \leq r \leq R\}, \quad h_j^*(0) = v_j, \quad j = 0, 1. \quad (11.2)$$

The machine data $\mathcal{M}$ include end caps Z_b, Z_t , rim levels H_j , phase volumes V_ℓ , densities ρ_ℓ , sound speeds c_ℓ , surface tensions σ_j , gravity g , acoustic frequency ω , source supports and boundary admittance Y . These are not silently changed to accommodate a target. The optical wavelength and acoustic frequency are different physical inputs.

The resulting Cartesian target is now supplied to the general variational inverse of Sections 2 and 4. The source optimization does not change this optical prescription.

12 Constructing the two prescribed Cartesian surfaces

12.1 Signed optical path and branch selection

For one surface, let $O = z_o$, $I = z_i$ be its laboratory conjugates, v its vertex, and n_o, n_i its indices. Define

$$D_o(r, z) = \sqrt{r^2 + (z - O)^2}, \quad D_i(r, z) = \sqrt{r^2 + (z - I)^2}, \quad (12.1)$$

$$F(r, z) = s_o n_o D_o + s_i n_i D_i - C, \quad C = s_o n_o |v - O| + s_i n_i |v - I|. \quad (12.2)$$

For finite nonzero conjugates and positive axial propagation at the vertex, $s_o = -\text{sgn}(O - v)$ and $s_i = \text{sgn}(I - v)$. The graph is the connected regular branch $F(r, h(r)) = 0$ through $h(0) = v$. This is the unsquared Cartesian construction of Silva-Lora and Torres [7]. A polynomial obtained by squaring may have additional roots and does not replace the branch test.

Let $\mathbf{x}$ be a surface point and set

$$\mathbf{d}_o = s_o \frac{\mathbf{x} - \mathbf{O}}{D_o}, \quad \mathbf{d}_i = s_i \frac{\mathbf{I} - \mathbf{x}}{D_i}, \quad \nabla F = n_o \mathbf{d}_o - n_i \mathbf{d}_i. \quad (12.3)$$

Since F is constant on the surface, its tangential gradient vanishes. For $\mathbf{T} = \mathbf{I} - \mathbf{m}\mathbf{m}^T$ and the normal $\mathbf{m}$ directed into the transmitted medium, this gives

$$n_o \mathbf{T} \mathbf{d}_o = n_i \mathbf{T} \mathbf{d}_i. \quad (12.4)$$

To select physical transmission rather than reflection or a wrong branch, put $\eta_n = n_o/n_i$ and $c_o = \mathbf{d}_o \cdot \mathbf{m}$ and evaluate

$$\mathbf{d}_t = \eta_n \mathbf{d}_o + \left[\sqrt{1 - \eta_n^2 (1 - c_o^2)} - \eta_n c_o \right] \mathbf{m}. \quad (12.5)$$

Require $c_o > 0$, a nonnegative radicand, forward interception of the next surface, and $\mathbf{d}_t = \mathbf{d}_i$. A finite margin from grazing and critical refraction is required for well-conditioned sensitivities.

12.2 Slopes and curvature without fitting a conic

Implicit differentiation supplies the exact graph derivatives:

$$h' = -\frac{F_r}{F_z}, \quad (12.6)$$

$$h'' = -\frac{F_{rr} + 2F_{rz}h' + F_{zz}(h')^2}{F_z}. \quad (12.7)$$

These require $F_z \neq 0$. For each distance $D = \sqrt{r^2 + (z - Z)^2}$,

$$D_{rr} = \frac{(z - Z)^2}{D^3}, \quad D_{rz} = -\frac{r(z - Z)}{D^3}, \quad D_{zz} = \frac{r^2}{D^3}. \quad (12.8)$$

The derivatives of F are the corresponding signed, index-weighted sums. At the vertex, $F_r = 0$ and $F_z = n_o - n_i$. Thus, for unequal indices,

$$h''(0) = \frac{n_i/(I - v) - n_o/(O - v)}{n_i - n_o}. \quad (12.9)$$

This is a useful independent paraxial check on the implicit surface, not an approximation used in its full-aperture construction. Equal indices, a focus at the vertex, collimated limits and graph folds require separate handling.

12.3 Shared intermediate conjugate and finite thickness

Apply the construction to

$$(n_o, z_o, n_i, z_i)_0 = (n_0, z_0 - v_0, n_1, z_1 - v_0), \quad (n_o, z_o, n_i, z_i)_1 = (n_1, z_1 - v_1, n_2, z_2 - v_1). \quad (12.10)$$

The common point is $\mathbf{Z}_1 = (0, 0, z_1)$, not equal vertex-relative distances. Consider the important branch with a real object before the lens, z_1 beyond both faces and a real final image after it. Let $\mathbf{x}_0$ and $\mathbf{x}_1$ be the actual successive ray intersections. Then

$$n_0 |\mathbf{x}_0 - \mathbf{Z}_0| + n_1 |\mathbf{Z}_1 - \mathbf{x}_0| = C_0, \quad (12.11)$$

$$-n_1 |\mathbf{Z}_1 - \mathbf{x}_1| + n_2 |\mathbf{Z}_2 - \mathbf{x}_1| = C_1. \quad (12.12)$$

Because the ray meets both faces before its intermediate focus,

$$|\mathbf{Z}_1 - \mathbf{x}_0| - |\mathbf{Z}_1 - \mathbf{x}_1| = |\mathbf{x}_1 - \mathbf{x}_0|. \quad (12.13)$$

Adding the two path equations gives the actual constant optical path

$$n_0|\mathbf{x}_0 - \mathbf{Z}_0| + n_1|\mathbf{x}_1 - \mathbf{x}_0| + n_2|\mathbf{Z}_2 - \mathbf{x}_1| = C_0 + C_1. \quad (12.14)$$

Together with the transmission checks, this proves axial geometrical stigmatism for this branch. Other real/virtual placements require the appropriate signed propagation segments. No aplanatic, achromatic or diffraction-limited conclusion follows from this identity alone.

13 Admissible geometry and compatibility with one machine

For full-frame graphs, define

$$V_0[h] = \int_{D_R} (h_0 - Z_b) \, dA, \quad V_1[h] = \int_{D_R} (h_1 - h_0) \, dA, \quad (13.1)$$

$$V_2[h] = \int_{D_R} (Z_t - h_1) \, dA, \quad V_0 + V_1 + V_2 = \pi R^2 (Z_t - Z_b). \quad (13.2)$$

The admissible set has prescribed V_0, V_1 , rim values $h_j(R) = H_j$, axis regularity $h'_j(0) = 0$ and positive gaps

$$h_0 - Z_b \geq d_{\min}, \quad h_1 - h_0 \geq d_{\min}, \quad Z_t - h_1 \geq d_{\min} > 0. \quad (13.3)$$

Volume preservation implies $\int \eta_0 \, dA = \int \eta_1 \, dA = 0$; the rim variations vanish. For a target, the measured compatibility residual is

$$c(\mathcal{P}; \mathcal{M}) = (h_0^*(R) - H_0, h_1^*(R) - H_1, V_0[h^*] - V_0^{\mathcal{M}}, V_1[h^*] - V_1^{\mathcal{M}}). \quad (13.4)$$

Its components have different units and need their own tolerances. An incompatible target is rejected, not altered by the acoustic optimizer. If Cartesian geometry is imposed only for $r \leq a < R$, full annular graphs must be explicitly supplied before these equations can be evaluated.

The family accessible to a sealed apparatus is contained in

$$\mathcal{C}_{\mathcal{M}} = \{\mathcal{P} : c(\mathcal{P}; \mathcal{M}) = 0, \text{ branch, gap and optical-access conditions hold}\}. \quad (13.5)$$

Consequently, arbitrary changes in conjugates and thickness need not be compatible even before acoustic reachability is considered. Where c is smooth and its Jacobian has constant rank r_c , the regular level-set theorem gives local dimension $\dim \mathcal{P} - r_c$ for the equality-constrained family. This is a conditional geometric statement, not an assumed rank count.

14 Verification obligations and scope of the result

The equations define a reusable inverse procedure, not universal feasibility. Before accepting a stationary SSL, verify the independent optical construction, volume/rim compatibility, full force residual, fixed-detector ray coverage, and fixed-command spatial convergence. Analytical checks should include:

1. The implicit vertex curvature in Eq. (12.9), and a supported collimated branch against its exact conic limit.

2. Planar hydrostatics, spherical Laplace pressure and the signed gravitational Hessian in Eqs. (5.5)–(5.8).
3. A layered analytic acoustic problem, transmission flux continuity and the independent power identity in Eq. (6.6).
4. Analytic source and shape derivatives against independent finite differences at multiple scales; physical and numerical conditioning reported separately.
5. Wave, surface and ray refinement at fixed commands, with the resulting numerical uncertainty compared against Eq. (9.10).

The report's existing 0.513 micrometre stationary candidate and partial formation diagnostic remain evidence for their recorded models only. No new simulation is performed by these derivations. A legitimate stationary claim is: *the specified commands realize the requested SSL within a declared, independently verified equilibrium model and tolerance*. Formation, stability and dynamic tracking are additional, separately testable claims.

Part II

Vision and software contract

Design proposal, 23 September 2026. **SSL** is the project's working name for a stigmatic singlet lens formed by two Cartesian surfaces. This document records the intended software and research contract, not an implemented API or a new experimental result. The accompanying [theory sketch](#) connects it to the general continuum formulation in [the canonical manuscript](#).

15 The objective

Given a fully specified SSL and a declared acoustic apparatus, find a bounded set of idealized source amplitudes and phases whose computed acoustic field supports that lens at equilibrium. Return a reproducible verification report, including unsuccessful outcomes. Do not change the requested optical system to obtain a more attractive result.

The immediate product is a **stationary lens compiler**:

```
Fixed lens prescription + fixed machine model
      ↓
Optical construction and apparatus compatibility
      ↓
Required full-interface mechanical loading
      ↓
Source amplitudes and phases
      ↓
Held-command coupled equilibrium and optical verification
      ↓
Versioned lens state with explicit evidence status
```

A later controller can form these states or move between them. A successful formation movie is not a prerequisite for a verified stationary solution. Conversely, stationary balance alone is not a formation or stability result. Idealized emitters are an intentional model choice; electrical calibration is not a version-1 acceptance requirement.

16 The user fixes the optical prescription

The version-1 input is

```
(n0, n1, n2, z0, z1, z2, front_vertex_z, central_thickness, frame_radius)
```

together with the design wavelength, clear aperture, branch convention and accuracy requirements. Distances in files and solvers are metres.

Input	Meaning and convention
<code>z0_m, z1_m, z2_m</code>	Object, shared intermediate conjugate, and final image, all in one laboratory coordinate system

Input	Meaning and convention
<code>front_vertex_z_m</code>	Axial position of the first surface vertex; it can be set to zero by an explicit coordinate convention
<code>central_thickness_m</code>	Positive vertex-to-vertex separation, not the separation of the two rim attachment planes
<code>frame_radius_m</code>	Bounding radius R at which the complete interfaces meet their supports
<code>clear_radius_m</code>	Qualified optical radius a , with $0 < a \leq R$; it is not silently reduced after ray tracing
<code>indices</code>	Three homogeneous refractive indices at the stated optical wavelength and material state
<code>wavelength_m</code>	Design wavelength; specifying it does not imply a diffraction calculation
<code>surface_domain</code>	Full-frame Cartesian surfaces or explicitly completed Cartesian pupils

The second vertex is `front_vertex_z_m + central_thickness_m`. Thus thickness must be included when translating the common intermediate point into the second surface's local coordinates. The user supplies `z1`; the compiler does not optimize it. It also leaves the conjugates, vertices, thickness, aperture and indices unchanged.

Special inputs such as collimated incidence need an explicit representation and analytic limiting branch, rather than an extremely large arbitrary distance. Equal indices, a conjugate at a vertex, and graph-folding branches must be diagnosed separately. The first implementation may reject unsupported cases with a precise reason rather than promise every real/virtual geometry.

16.1 What “the surfaces are known” means

For **full-frame Cartesian surfaces**, the supplied prescription selects two regular, physically transmitting branches over $0 \leq r \leq R$. Their entire shapes, edge heights and enclosed lens volume follow from that prescription.

For **Cartesian pupils with non-optical annuli**, the prescription determines only $0 \leq r \leq a$. The annular profiles on $a < r \leq R$ must also be provided or generated by an explicit, deterministic completion rule with its mounting, smoothness and volume data. They cannot be ignored in the mechanics or wave problem. Version 1 does not secretly optimize those annuli.

Proposed default, requiring confirmation before API implementation: use full-frame Cartesian surfaces, with a separately declared clear aperture. Retain pupil-plus-annulus mode as an explicit option because existing runs use it. The two modes are not interchangeable prescriptions.

17 The machine is a separate input

The machine model declares the cylinder and end-cap positions, source support and spatial patterns, active/inactive regions, optical access, acoustic boundary admittances, phase materials, gravity, attachment conditions, and source-command limits. Frequency is fixed for a version-1 compilation; an optional frequency study consists of separately recorded cases.

Material data must distinguish supplied values, engineering assumptions and measured quantities. A stationary inviscid acoustic calculation does not use viscosity merely because viscosity is present in a material file. Immiscibility, constant surface tension and perfect pinning are also declared assumptions.

Two operating modes must remain distinct:

- **Design a machine/fill for one lens:** derive its required rim heights and volumes, then test a declared compatible apparatus. This is not evidence that an existing machine can change to that lens without modification.
- **Compile for a fixed machine:** treat all geometry, phase volumes and attachment data as immutable. Reject incompatible prescriptions before source optimization. Do not add fluid, move a rim or change an acoustic window without declaring a different apparatus.

For a sealed reconfigurable machine, volume and rim compatibility are central. Changing z_0 , z_1 , z_2 or thickness generally changes the complete Cartesian profiles and their volume. Therefore the set of SSLs available to one machine is a constrained family, not automatically an arbitrary box of optical inputs. An explicitly designed non-optical annulus may supply accommodation; pumps or moving supports would instead be additional apparatus components and control variables, outside this fixed-machine first version.

18 Responsibilities of the compiler

18.1 A. Optical construction

Construct each unsquared constant-optical-path branch through its prescribed vertex. Check graph regularity, orientation, Snell transmission, ray interception, minimum thickness and clearance to the cylinder boundaries. Trace the ideal pair to verify the common-conjugate construction independently of emitter synthesis. A Cartesian polynomial root is not sufficient branch verification.

18.2 B. Required loading

Calculate nonlinear capillary and hydrostatic loading over both complete interfaces, modulo the constant pressure multipliers imposed by sealed volumes. Keep the signed density difference on each interface. Store the geometry, load, units and sign convention together.

18.3 C. Source inversion

At the prescribed geometry, construct the acoustic response to the declared source space. Optimize complex commands under explicit amplitude and effort constraints, retaining coherent interference and forces on both whole faces. Multiple starting points and alternative discretizations are numerical tools, not changes to the optical prescription. A local optimizer's failure is not proof that a requested lens is unreachable.

18.4 D. Held-command verification

Freeze the physical command distribution. Solve the actual coupled acoustic and mechanical equations, then trace the resulting surfaces at the fixed detector and fixed illumination. A root solver may start near the target; that is a legitimate equilibrium search, not a physical trajectory.

Refine acoustic and surface discretizations with the same source regions, spatial command patterns, amplitudes, phases and frequency. Increasing the emitter count or re-optimizing produces a new design, not a convergence test. Report numerical uncertainty relative to the requested tolerances.

19 Acceptance without ambiguous success labels

Use separate evidence fields rather than one “success” flag:

Evidence field	What it means
<code>optically_constructed</code>	The prescribed ideal pair passes branch and optical checks
<code>apparatus_compatible</code>	Shapes, supports, phase volumes and access match the declared machine
<code>source_candidate_found</code>	An inverse solve found admissible complex commands
<code>stationary_root_converged</code>	The full discrete coupled force residual meets its tolerance
<code>stationary_optical_gates_met</code>	Both height errors, maximum spot and ray coverage pass
<code>fixed_command_convergence_verified</code>	Independent refinements support the reported stationary metrics
<code>stability_assessed</code>	A specified perturbation model has actually been analysed; include its conclusion and scope
<code>formation_verified</code>	A specified startup trajectory reaches and maintains the accepted operating state
<code>transition_verified</code>	A specified change between operating states meets its transient requirements

Unknown, failed and not-applicable are distinct. “Unreachable” is reserved for a valid necessary-condition obstruction or other certificate within the declared model; an exhausted optimization budget returns “no solution found.”

Height error is the maximum absolute cycle-mean lab-coordinate departure from each prescribed surface on its qualified pupil, without fitted piston, tilt or focus. Optical error is maximum geometric radius about the prescribed axis at the fixed detector, with a separate full-coverage requirement. Neither an RMS radius nor the radius of a clipped surviving subset substitutes for these gates. Sampling resolution and any non-certified continuous maximum must be stated.

The intended scales are 10 nm height and approximately 0.5 μm spot. Executable requests need exact numbers and inequalities: a strict 0.500 μm bound would **not** accept the saved 0.513 μm result. User acceptance of an approximate spot does not silently relax the independent height or coverage criteria.

20 Proposed software interfaces and records

The following names are a proposal, not existing callable functions:

```
target = construct_ssl(prescription)
compatibility = check_machine(target, machine)
load = required_load(target, machine)
candidate = synthesize_sources(target, load, machine, inverse_settings)
equilibrium = solve_fixed_command(candidate, machine)
evidence = verify_stationary(target, equilibrium, candidate, checks)
```

Keep immutable prescription, machine, target, source-command, equilibrium and verification objects separate. Cache keys must include geometry, materials, boundary operators, source patterns,

frequency and numerical settings. A response matrix for one target cannot silently be reused after changing its scattering geometry.

Each result belongs in a protected run directory containing requested and resolved configurations; target and computed surfaces; complex commands and source-region geometry; optical and force metrics; source/provenance hashes; mesh and solver histories; and a human-readable report. A stationary state has no physical time axis. A trajectory adds actual times, command history, fluid states and per-frame optical checks.

The notebook is a reader and verification presentation, not an independent source of physical states. Display both unforced and driven surfaces, source geometry, harmonic pressure and mean traction separately, and the full spot diagram. Formation and carrier-wave animations have different time scales and must never be spliced into one implied successful experiment.

21 Extension to an adaptive SSL machine

Once stationary states are independently verified, a lens library maps valid prescriptions to commands and evidence. The first adaptive mode can be **select, settle, then image**. A later mode tracks a time-dependent optical request with bounded transient error.

Endpoint existence does not establish a connecting path. Changes must preserve the apparatus constraints, respect actuator bandwidth, and avoid branch loss, interface contact and unacceptable tracking error. Command interpolation is at most an initial guess for a transition calculation. Stability and formation may require feedback; specify the observations and controller rather than assuming exact knowledge of the liquid surface.

Axial stigmatism is defined for the selected conjugate pair and optical indices. It does not automatically provide aplanatism, broadband correction, a large corrected field or a zero diffraction spot. UV curing would create a different manufacturing workflow, with evolving shape and refractive index; it is not part of continuously reconfigurable liquid operation.

22 What exists and what should be implemented next

Existing code supplies Cartesian targets, idealized source regions, quadratic force synthesis, held-command equilibrium checks, ray tracing and provenance. The unified fixed-prescription API and its compatibility contract are still proposed. In particular, the current two-face surface basis uses fixed rim levels and fixed volume relative to reference planes. It must not be treated as a general full-frame Cartesian-volume implementation without extension.

The [current stationary notebook](#) records a 0.513 μm surviving-ray spot, 20.2/24.4 nm height errors, and one failed ray out of 4001; fixed-command refinement is unfinished. The [formation notebook](#) is a separate unsuccessful, partial Stokes diagnostic. Neither demonstrates the proposed adaptive family.

The next implementation milestones are:

1. Fix the prescription schema, coordinate conventions and explicit full-frame versus pupil-completion mode; test optical and volume compatibility.
2. Connect the existing solvers through immutable inputs and explicit evidence fields, without changing prescribed geometry during source fitting.
3. Verify a small, independently selected family with fixed-command refinement, including unsuccessful or incompatible cases, before mapping a wider range.
4. Add stability-qualified operation and then formation/transition control.

The research claim to pursue is a reproducible map from **specified compatible SSLs to verified acoustic commands**, with quantified errors, effort and limitations—not universal realizability inferred from a single lens.

23 Scientific grounding

The optical construction follows Silva-Lora and Torres, Superconical aplanatic ovoid singlet lenses [7] (2020). Its two-surface construction is an optical foundation, not evidence that this acoustic machine realizes the surfaces. The acoustic/mechanical inverse and its assumptions are developed separately in the [theory sketch](#) and canonical manuscript.

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